

FINAL PROJECT DOCUMENTATION

Title: Analyse OTT Platform Viewing Patterns to Support Data-Driven Content Decisions (MediaTech)

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Program: Professional Program in Data Analytics with GenAI

Course: Applied Data Analytics Capstone Project

Institution: NIIT

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1. Project Overview

This project focuses on analysing OTT viewing patterns, user engagement, content performance and subscription retention behaviour to support data-driven business decisions.

The project is based on the **MediaTech OTT case study**, covering the OTT platforms **NetStream, AmazePrime and DizPlay+**. The analysis integrates five datasets containing user profiles, content information, viewing activity, subscription and retention information, and ratings and feedback.

The primary objective is to identify meaningful patterns in viewer behaviour and content performance and convert these findings into actionable recommendations for **content acquisition, content curation, viewer engagement and customer retention**.

The project follows the NIIT-defined analytics workflow and uses Excel, Python, SQL, Power BI, statistical analysis and Simple Linear Regression to address the identified business questions.

2. Data Sources

The analysis uses five datasets provided as part of the NIIT MediaTech OTT case study:

1.  **User Profile**
2.  **Content Library**
3.  **Viewing Activity**
4.  **Subscription & Retention**

5. ★ Ratings & Feedback

These datasets were integrated conceptually through common identifiers such as **User_ID** and **Content_ID** to analyse relationships between user characteristics, content, viewing behaviour, feedback and subscription outcomes.

3. Project Objectives

The project aims to achieve the following objectives:

Identify High-Engagement Content

Identify high-performing genres, languages, content types and OTT platforms to support content acquisition and curation decisions.

Understand Viewing Behaviour

Analyse viewing patterns across user demographics, regions, devices, platforms and time periods.

Analyse Subscription & Retention

Examine subscription renewal and churn behaviour and identify engagement-related patterns associated with retention.

Evaluate Content Success

Analyse views, watch duration, completion percentage, ratings and feedback to understand content performance.

Support Data-Driven Decisions

Convert analytical findings into actionable recommendations for content strategy, viewer engagement and customer retention.

4. Tools & Technologies

Tool / Technology	Application
Microsoft Excel	Data analysis, formulas, PivotTables, KPIs, outlier analysis and dashboard development

Tool / Technology	Application
Python	Data preprocessing, exploratory data analysis and statistical exploration
Pandas & NumPy	Data manipulation and numerical analysis
Matplotlib & Seaborn	Data visualisation
SQL / MySQL	Data querying, joins, aggregations and business analysis
Microsoft Power BI	Interactive dashboard, data modelling and reporting
DAX	KPI and calculated measure development
Jupyter Notebook / Google Colab	Python analysis and documentation

5. Methodology & NIIT Project Workflow

The project follows the **11-phase NIIT analytics workflow**.

Phase 1: Kick-off & Project Allocation

Understood the MediaTech OTT case study, project objectives, expected deliverables and analytical scope.

Phase 2: Data Familiarisation, Cleaning & Version Control

Profiled the five datasets, examined data types and quality, cleaned and standardised the datasets, and maintained separate cleaned versions for analysis.

Phase 3: Business Problem & Hypothesis Framing

Defined the business problem, project objectives, key business questions and hypotheses related to viewing behaviour, completion, retention and churn.

Phase 4: Excel Analysis & Dashboard

Performed formula-based analysis, XLOOKUP-based data enrichment, PivotTable analysis, KPI calculations, outlier analysis and developed the Excel dashboard.

Phase 5: Python EDA

Performed exploratory data analysis using Pandas, NumPy, Matplotlib and Seaborn to identify distributions, trends, relationships and patterns.

Phase 6: SQL Analysis

Loaded the datasets into MySQL and performed data-quality checks, joins, aggregations, subqueries, derived-table analysis, window functions and business-question analysis.

Phase 7: Power BI Dashboard

Developed an interactive Power BI dashboard using data modelling, DAX measures, KPIs, filters and visualisations.

Phase 8: Statistical Validation

Applied probability, Z-tests, T-tests and Chi-Square tests to evaluate selected business hypotheses and determine whether observed patterns were statistically significant.

Phase 9: Simple Linear Regression

Developed a Simple Linear Regression model to examine whether **Watch Duration** can predict **Completion Percentage**. The model was evaluated using R^2 and error metrics.

Phase 10: Business Insights, Recommendations & Future Scope

Converted analytical findings into business insights, recommendations, limitations and opportunities for future development.

Phase 11: Final Documentation & Presentation

Consolidated the complete project work into final documentation and presentation deliverables.

6. Dataset Understanding & Data Dictionary

The project uses five datasets covering users, content, viewing activity, subscriptions and feedback.

Dataset	Records	Key Field	Purpose
 User Profile	2,000	User_ID	User demographics and subscription profile
 Content Library	17,797	show_id / Content_ID	Content and metadata information

Dataset	Records	Key Field	Purpose
 Viewing Activity	30,000	Session_ID	Viewing and engagement behaviour
 Subscription & Retention	2,000	User_ID	Subscription renewal and churn information
 Ratings & Feedback	12,000	User_ID + Content_ID	Ratings, likes and user feedback

Key Relationships

- **User_ID** connects user profile, viewing activity, subscription and retention information.
- **Content_ID** connects viewing and feedback data with the content library.
- Content identifiers were standardised during preprocessing to establish consistent relationships.
- A unique **Viewing_Record_ID** was created during preprocessing to distinguish viewing records where duplicate Session_ID values were identified.

These relationships supported integrated analysis across user, content, viewing and retention dimensions.

7. Data Preprocessing & Cleaning

The datasets were profiled and cleaned before analysis to improve consistency and analytical reliability.

Key Preprocessing Steps

- Checked missing values and duplicate records.
- Reviewed data types and numerical ranges.
- Standardised text fields and categorical values.
- Corrected and standardised Content ID formats.
- Recovered incorrectly stored duration values in the Content Library where possible.
- Flagged invalid or zero-duration records for analytical review.
- Investigated duplicate Session_ID values and created a unique Viewing_Record_ID.
- Validated dates, numerical values and key relationships.
- Reviewed potential outliers, particularly in Watch Duration.

- Retained questionable records where deletion could introduce unsupported assumptions.

The cleaned datasets were subsequently prepared for **Excel, Python, SQL and Power BI analysis**.

8. Business Problem Statement

Business Problem

“Analyse OTT platform viewing behaviour, content performance, user characteristics, ratings and subscription retention to identify factors influencing engagement and churn and support data-driven content, platform and customer-retention decisions.”

The OTT platform needs to understand how users interact with content and platforms and how viewing behaviour relates to subscription retention.

The key business challenge is to identify patterns associated with **content success, viewer engagement, subscription renewal and churn**, enabling the organisation to make better content and customer-retention decisions.

Key Business Questions

1. Content Performance

Which genres, languages and OTT platforms have the highest viewer engagement and completion rates?

2. User Viewing Behaviour

How do viewing patterns differ across user demographics, regions, devices and time of day?

3. Content Success & Feedback

How are ratings, likes and user feedback related to content completion and rewatch behaviour?

4. Subscription Retention & Churn

What user engagement and subscription factors are associated with subscription renewal and churn?

5. Viewing Engagement & Completion

Does Watch Duration significantly influence Completion Percentage, and can it be used to predict completion?

9. Statistical Validation & Simple Linear Regression & Model Evaluation

9.1 Statistical Validation

Probability and hypothesis testing were performed to validate selected OTT business questions.

The analysis included **Z-tests, T-tests and Chi-Square tests** to examine churn, renewal, viewing behaviour and categorical associations.

Key Statistical Analyses

-  **Probability:** Churn probability.
-  **Z-Test:** Churn and renewal proportions against selected benchmarks.
-  **T-Test:** Comparison of average Watch Duration between churned and non-churned users.
-  **Chi-Square Test:** Association between selected categorical variables such as subscription, renewal, device type and churn.

Test statistics and p-values were evaluated, followed by **Reject H_0 / Fail to Reject H_0** decisions and business interpretation.

Key Churn Result

There were **888 churned users out of 2,000 users**, resulting in an overall churn rate of approximately **44.4%**.

The churn proportion was significantly higher than the selected benchmark of 20%, supporting the conclusion that customer churn is a significant business concern in the analysed dataset.

9.2 Simple Linear Regression

A Simple Linear Regression model was developed to examine whether **Watch Duration** can predict **Completion Percentage**.

The analysis used Watch Duration as the independent variable and Completion Percentage as the dependent variable. The data was divided into **training, validation and**

testing sets, predictions were generated, and model performance was evaluated using relevant regression metrics including **R², MAE, MSE and RMSE**.

Key Regression Result

Correlation (R): 0.2205

R²: 4.86%

p < 0.001

The analysis indicates a **statistically significant but weak positive relationship** between Watch Duration and Completion Percentage.

The R² value of approximately **4.86%** indicates that Watch Duration explains only a small proportion of the variation in Completion Percentage. Therefore, Watch Duration alone has **limited predictive power**, and other factors should be considered for a stronger completion prediction model.

9.3 Model Analysis – Simple Linear Regression

Simple Linear Regression was applied to analyse whether **Watch Duration** can predict **Completion Percentage**. Model performance was evaluated using **R², MAE, MSE, and RMSE**. The model showed a weak positive relationship, indicating that watch duration alone has limited predictive power for completion percentage.

10. Excel Analysis

Excel was used for data validation, exploratory analysis, KPI calculation, PivotTables, outlier analysis and dashboard preparation.

The cleaned datasets were structured into Excel tables and analysed using formulas including **XLOOKUP, IF and IFERROR**, along with conditional formatting, PivotTables and PivotCharts.

Key Excel Analysis Performed

- Analysed viewing activity by genre, content type, platform, region, language, device and time of day.
- Compared Movies and TV Shows based on viewing volume, watch duration and engagement.
- Identified high-performing genres and titles using viewing and engagement measures.
- Analysed subscription type, renewal and churn behaviour.

- Examined ratings, completion percentage and rewatch-related behaviour.
- Created derived fields such as Viewing Year, Content Duration, Content Age and engagement-related categories.
- Performed Watch Duration outlier analysis using the IQR method.
- Created PivotTables and supporting charts for dashboard development.

Outlier Analysis

The IQR method produced an upper statistical boundary of approximately **247.13 minutes** for Watch Duration.

Separately, a **business-defined threshold of greater than 225 minutes** was used to identify unusually long viewing sessions for business analysis. These thresholds were treated as separate analytical criteria.

Key Excel Findings

- TV Shows generated substantially higher total watch hours and average viewing duration than Movies, indicating stronger long-form viewing engagement.
- Drama recorded the highest viewing volume among the analysed genres.
- Viewing behaviour varied across platforms, regions, devices and time periods.
- Overall churn was approximately **44.4%**, indicating a significant retention challenge.
- Watch Duration and Completion Percentage showed a weak positive relationship.

The Excel analysis provided the foundation for the subsequent Python, SQL and dashboard analysis.

11. Python EDA

Python was used to perform detailed Exploratory Data Analysis, numerical analysis, visualisation and statistical exploration.

The cleaned datasets were analysed using **Pandas and NumPy**, while **Matplotlib and Seaborn** were used to create visualisations.

Major EDA Activities

- Inspected dataset structure, dimensions and data types.
- Checked missing values and duplicate records.
- Examined numerical distributions using descriptive statistics.
- Analysed categorical variables including genre, platform, content type, subscription type, region and device.

- Investigated viewing behaviour and engagement patterns.
- Analysed churn and retention across user and subscription characteristics.
- Examined ratings, completion percentage and Watch Duration.
- Performed correlation analysis between relevant numerical variables.
- Investigated potential outliers in viewing duration.
- Created histograms, boxplots, bar charts, count plots, heatmaps and relationship plots.

Important EDA Findings

The correlation between **Watch Duration and Completion Percentage** was **approximately 0.2205**, indicating a weak positive relationship.

The relationship between **Content Duration and Completion Percentage** was **approximately -0.032**, indicating an extremely weak negative relationship.

The EDA also showed that TV Shows generated substantially greater watch-hour engagement than Movies, while viewing behaviour differed across genres and platforms.

Python EDA helped validate the patterns identified through Excel and provided additional evidence for the statistical validation and regression stages.

12. SQL Analysis

MySQL was used to perform structured relational analysis and answer the major business questions.

The five datasets were loaded into the **ott_analysis** database using the following tables:

- user_profile
- content_info
- viewing_activity
- subscription_retention
- ratings_feedback

SQL Analysis Performed

The SQL analysis included:

- Data quality and completeness checks.
- Duplicate and identifier validation.
- Referential integrity checks.
- Descriptive statistics and aggregation.

- Median and percentile-related analysis.
- INNER JOIN and LEFT JOIN operations.
- Subqueries and derived tables.
- GROUP BY and HAVING analysis.
- Conditional aggregation.
- Window functions for ranking and comparative analysis.
- Temporary tables for intermediate analysis.
- Views for reusable analytical datasets.
- Stored procedures for repeatable analytical tasks.
- Exception handling where implemented within procedures.
- Triggers for data-related validation and control.
- Business-question-driven queries connecting analytical results to business interpretation.

Business Areas Analysed

SQL was used to analyse:

- Content performance.
- Platform performance.
- Genre performance.
- Viewing engagement.
- Ratings and feedback.
- Subscription behaviour.
- Renewal and churn.
- Regional patterns.
- User-level activity.

The SQL analysis provided a structured and scalable approach to validating findings from Excel and Python and demonstrated the ability to translate business questions into SQL-based analysis.

13. Excel Dashboard

An interactive Excel dashboard was developed to provide a consolidated view of OTT viewing behaviour, content performance and subscription retention.

The dashboard uses **PivotTables, PivotCharts, slicers, calculated metrics and a consistent visual design** to support business users in exploring the data.

Dashboard Components

The dashboard includes:

- Churn by Subscription Type.
- Platform Performance.
- Viewing by Time of Day.
- Top 10 Genres.
- Movie vs TV Show Performance.
- Regional Churn.
- Top 10 Titles.
- Year-Month Viewing Trend.
- Key engagement and retention KPIs.

Dashboard Purpose

The dashboard enables decision-makers to identify:

- High-demand content categories.
- Platforms with stronger viewing engagement.
- Changes in viewing behaviour over time.
- Regions and subscription types with higher churn.
- High-performing titles and genres.
- Differences between Movies and TV Shows.

The Excel dashboard provides a business-friendly summary of the detailed analytical work performed during the project.

14. Power BI Dashboard & KPIs/DAX

Power BI was used to develop an interactive Business Intelligence dashboard for deeper analysis and decision support.

The data model was designed around user, content, viewing, ratings and subscription information. Relationships between relevant tables were established to support filtering and cross-analysis.

Key Power BI KPIs

The dashboard includes measures such as:

- Total Users.

- Total Viewing Records.
- Total Watch Hours.
- Average Watch Duration.
- Average Completion Percentage.
- Average Rating.
- Churned Users.
- Retained Users.
- Churn Rate.
- Renewal Rate.

DAX Analysis

DAX measures were created to calculate KPIs dynamically based on report filters and user selections.

The measures include calculations for:

- Total and distinct user counts.
- Total watch duration and watch hours.
- Average viewing duration.
- Average completion percentage.
- Churn and retention.
- Churn percentage.
- Renewal percentage.
- Filter-responsive performance measures.
- Time-based calculations using the Date table.

The Power BI dashboard provides interactive filtering and analysis by **platform, region, content type, genre, subscription type and time**.

Where implemented in the final Power BI report, **Row-Level Security (RLS)** was used to restrict regional access according to defined analyst roles.

15. Business Insights

The combined Excel, Python, SQL and Power BI analysis generated the following major business insights.

1. TV Shows Show Stronger Viewing Engagement

Movies generated a high number of viewing records, while TV Shows generated substantially higher total watch hours and average viewing duration. This indicates stronger long-form viewing engagement for TV Shows.

2. Drama Has Strong Viewing Demand

Drama recorded the highest viewing volume among the analysed genres, indicating strong audience demand and potential value for content acquisition and recommendation strategies.

3. Churn Is a Major Business Challenge

Approximately **44.4% of users were classified as churned**, while approximately **55.6% were retained**. This indicates a significant opportunity to improve customer retention.

4. Watch Duration Has Limited Predictive Power for Completion

Watch Duration and Completion Percentage had a correlation of approximately **0.2205**, indicating a weak positive relationship. Watch Duration alone is therefore insufficient to explain or accurately predict completion behaviour.

5. Content Duration Is Not a Strong Completion Driver

The correlation between Content Duration and Completion Percentage was approximately **-0.032**, indicating an extremely weak relationship.

6. Viewing Behaviour Varies Across User Segments

Viewing behaviour differs across platform, region, device, subscription type and time of day. These differences create opportunities for targeted content and engagement strategies.

7. Statistical Testing Supports the Retention Concern

The churn proportion was significantly higher than the selected benchmark, providing statistical evidence that churn represents an important business concern in the analysed dataset.

Overall, the findings indicate that **content strategy and customer-retention strategy should be considered together when making OTT business decisions**.

16. Recommendations

Based on the analytical findings, the following recommendations are proposed.

1. Strengthen TV Show Content Strategy

Increase focus on high-engagement TV Shows, particularly categories demonstrating strong viewing demand and watch-hour contribution.

2. Expand High-Performing Genres

Use viewing volume, watch hours, completion and ratings together to identify genres suitable for content acquisition and curation.

3. Improve Churn Prevention

Develop targeted retention campaigns for users showing declining engagement or other characteristics associated with churn.

4. Personalise Content Recommendations

Use viewing history, preferred genres, platform behaviour and ratings to provide more relevant content recommendations.

5. Segment Users

Create meaningful user segments based on subscription type, engagement level, region, device and viewing behaviour to support targeted marketing and retention initiatives.

6. Monitor Content Performance Continuously

Track views, watch hours, completion, ratings and rewatch behaviour to identify successful and underperforming content.

7. Use Multiple Metrics for Decision Making

Content decisions should consider multiple indicators rather than relying only on viewing volume. Combining engagement, completion, ratings and retention provides a more balanced view of content performance.

17. Limitations

The following limitations should be considered when interpreting the results:

- The analysis is based on the datasets provided in the NIIT MediaTech case study and may not represent a complete real-world OTT environment.
- Some records contained unusual or questionable values, particularly viewing-duration values. These were investigated and retained where removal could introduce unsupported assumptions.

- Correlation does not imply causation. Observed relationships should not automatically be interpreted as causal relationships.
- The Simple Linear Regression model uses only Watch Duration as the predictor of Completion Percentage.
- The regression model has limited explanatory power, with an R^2 of approximately **4.86%**.
- Other potentially important factors such as pricing, promotions, recommendation algorithms, content availability and external events were not available in the provided datasets.
- Dashboard results depend on the quality and completeness of the underlying data.
- Statistical conclusions are specific to the available dataset and selected hypotheses.

Therefore, the findings should be treated as **decision-support insights rather than definitive causal conclusions**.

18. Future Scope

The project can be extended to provide deeper analytical and business value.

Advanced Predictive Analytics

Future models could include:

- Logistic Regression for churn prediction.
- Decision Trees and Random Forests for user classification.
- Classification models for retention prediction.
- Clustering for viewer segmentation.
- Recommendation systems for personalised content suggestions.

Advanced Time-Series Analysis

Future work could forecast:

- Viewing demand.
- Watch hours.
- Subscriber growth.
- Churn trends.
- Content demand by genre.

Enhanced Customer Analytics

User-level behavioural features could be incorporated to develop early-warning systems for users who are likely to churn.

Recommendation Engine

A recommendation system could be developed using viewing history, ratings, genres, completion and user preferences.

Real-Time Analytics

Future implementation could use streaming data pipelines and real-time dashboards to monitor content performance and user engagement.

Additional Business Data

Future analysis could integrate:

- Subscription revenue.
- Marketing campaign data.
- Content acquisition cost.
- Content production cost.
- Advertisement revenue.
- Customer support interactions.

This would enable analysis of **content ROI, customer lifetime value and revenue-related performance** in addition to viewing behaviour.

19. Conclusion

This project analysed OTT platform viewing patterns to support data-driven content and customer-retention decisions.

The analysis integrated five datasets covering **user profiles, content information, viewing activity, subscription and retention, and ratings and feedback**. Excel, Python, SQL and Power BI were used for data preparation, exploratory analysis, statistical validation, regression analysis and interactive reporting.

The analysis identified several important patterns. **TV Shows demonstrated substantially stronger viewing engagement in terms of watch hours and average viewing duration**, while Drama showed strong viewing demand. At the same time, the overall churn rate of approximately **44.4%** highlights a significant customer-retention challenge.

Statistical analysis provided evidence supporting the importance of the churn problem. The Simple Linear Regression analysis showed a **weak positive relationship between Watch Duration and Completion Percentage**, with an R^2 of approximately **4.86%**. This indicates that Watch Duration alone has limited predictive power and that multiple behavioural and content factors should be considered.

The Excel and Power BI dashboards transform the detailed analytical results into accessible business intelligence outputs, enabling stakeholders to explore viewing, content and retention patterns interactively.

Overall, the project demonstrates how **data cleaning, exploratory analysis, SQL, statistical validation, regression, visualisation and Business Intelligence can be combined to transform OTT data into actionable business insights and support data-driven decision making**.

20. References / Appendix

References

The project was developed using the following resources:

1. **NIIT Professional Program in Data Analytics with GenAI** – course materials and Applied Data Analytics Capstone Project guidance.
2. **MediaTech OTT Case Study** – datasets and business problem provided as part of the NIIT capstone project.
3. **Microsoft Excel** – data cleaning, analysis, formulas, PivotTables, PivotCharts and dashboard development.
4. **Python** – Pandas, NumPy, Matplotlib and Seaborn for data analysis and visualisation.
5. **MySQL** – relational database analysis, joins, aggregations, subqueries, views, procedures, triggers and window functions.
6. **Microsoft Power BI** – data modelling, interactive dashboards and DAX-based KPI calculations.
7. **Jupyter Notebook / Google Colab** – Python-based exploratory analysis and documentation.

Appendix

The following project artefacts were created as part of the capstone project:

- Cleaned Excel datasets.
- Consolidated Excel analysis workbook.

- Excel dashboard.
- Python EDA notebook.
- MySQL database and SQL analysis script.
- Power BI dashboard.
- DAX measures and calculated KPIs.
- Statistical hypothesis-testing outputs.
- Simple Linear Regression analysis.
- Business insights and recommendations.
- Final project documentation.
- Project presentation.

Key Analytical Outputs

Dataset Size

Dataset	Records
User Profile	2,000
Content Library	17,797
Viewing Activity	30,000
Subscription & Retention	2,000
Ratings & Feedback	12,000

Key KPIs

KPI	Result
Churned Users	888
Retained Users	1,112
Churn Rate	44.4%
Average Watch Duration	≈ 181 minutes
Average Completion Percentage	≈ 52.36%
Watch Duration vs Completion Correlation	≈ 0.2205
Regression R ²	≈ 4.86%

These outputs provide supporting evidence for the conclusions, insights and recommendations presented in this documentation.

NIIT Workflow Coverage Summary

Project Outcome: The project provides an end-to-end analytics solution for understanding OTT viewing behaviour, content performance and customer retention, with the final outputs designed to support data-driven business decisions.